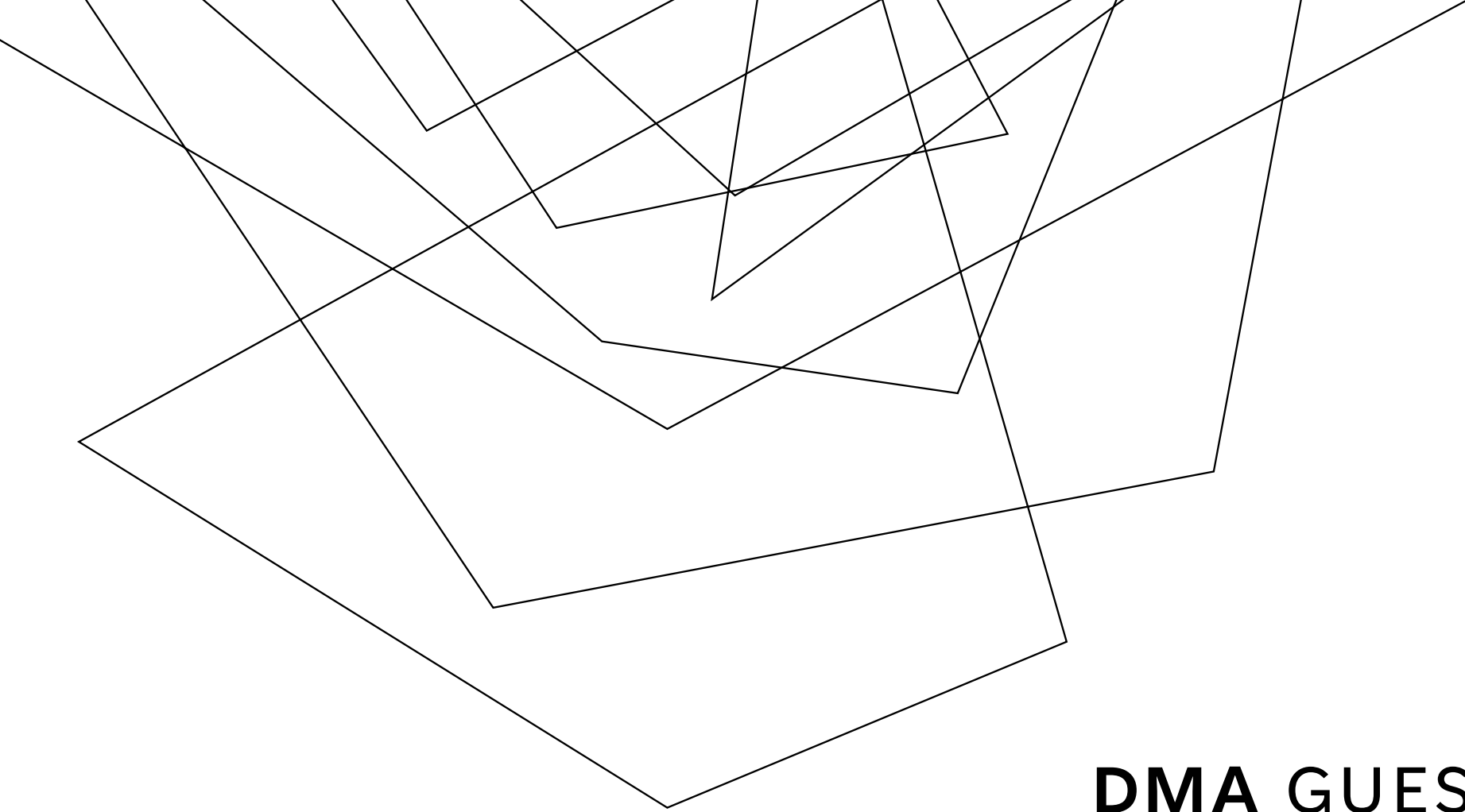




DMA GUEST LECTURE

BART SANTING

AGENDA

About me / my research

Generative AI: Theory

Neural networks, pre-training, supervised
fine-tuning, RLHF and bias

Generative AI: Practice

Questions



ABOUT ME & MY RESEARCH

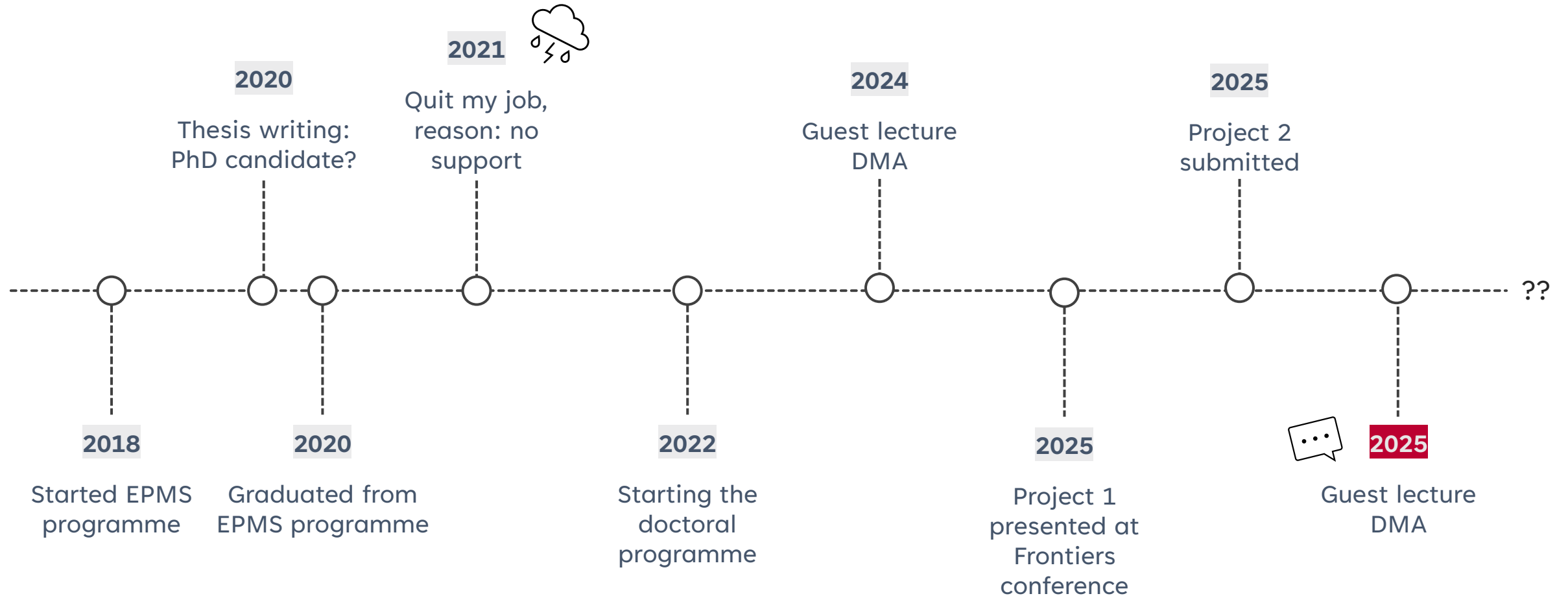
I WORK AS A DATA SCIENTIST FOR OHRA



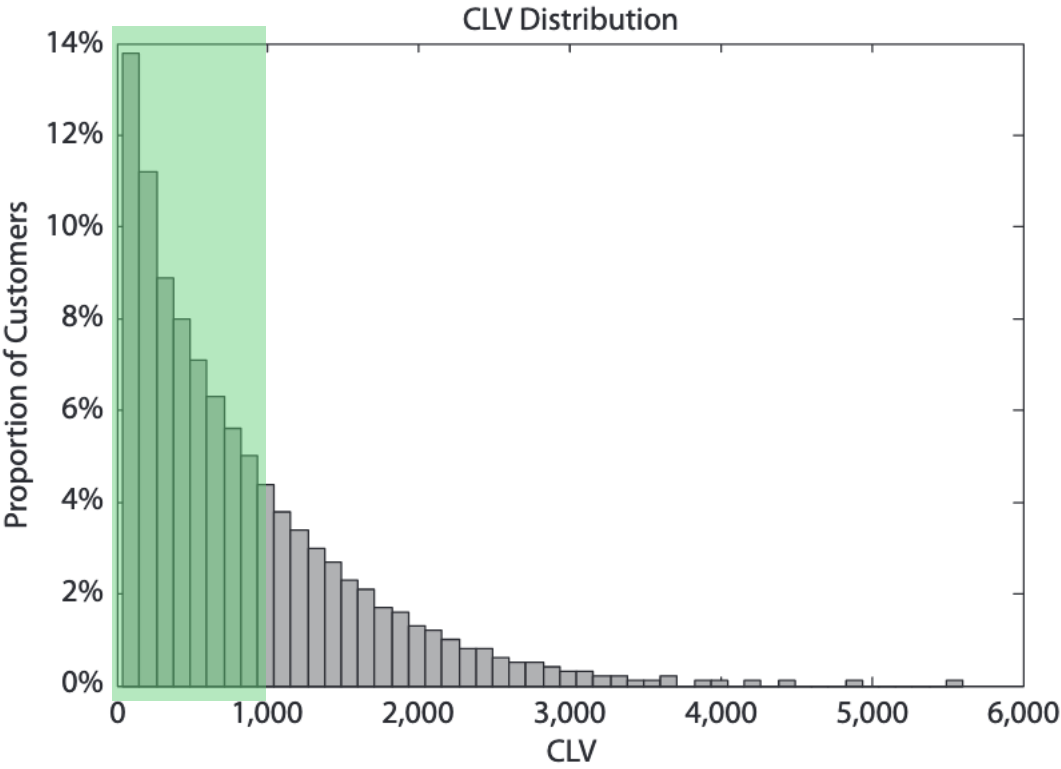
Active in 11 countries
16.000 coworkers
19.000.000 clients
10.700.000.000 turnover
Life and Non-life products
Partnerships with dozens of
parties all over the world.

Direct Writer
450 coworkers
900.000 clients
250.000.000 turnover
14 non-life products
Healthcare insurer
Partnerships with several
parties in the Netherlands

MY JOURNEY INTO ACADEMIA



PROJECT 1: SERVICE THAT STICKS



Typically, a large portion of an organization’s customer base is made up of lower-value customers¹.

		Targeted Customers	
		High Value	Low Value
Tactical Approach	Offense	Premium Offerings	Loyalty Programs
	Defense	Strategic Acct Management	Customer Service

Improving customer service is an effective tactical approach for low-value customers retention¹.

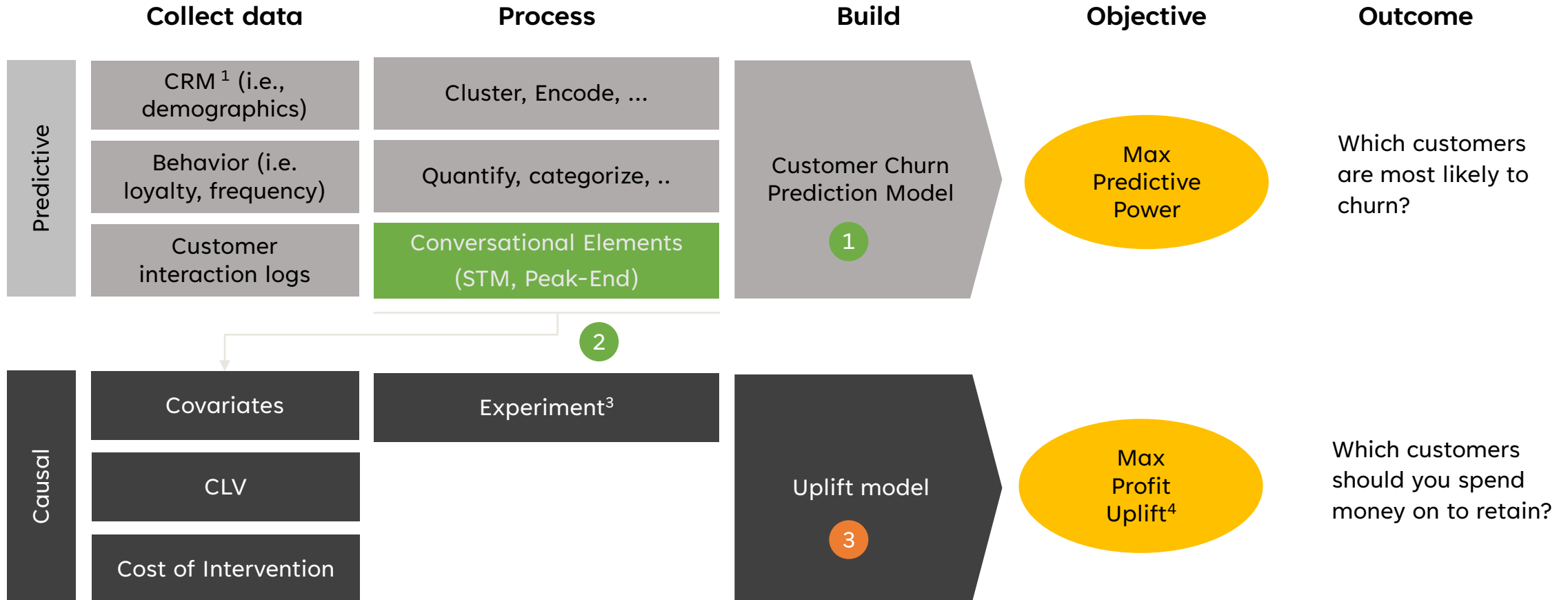
Table 1
Four theoretical classes.

Churn when targeted	Yes	Do-Not-Disturbs	Lost Causes
	No	Sure Things	Persuadables
		No Churn when not targeted	Yes

Precision is required to target only the persuadables within the customer base².

1 Fader, P., & Toms, S. E. (2018). Chapter 3 Using Customer Centricity to Tune Retention and Development Tactics. In *The Customer Centricity Playbook: Implement a Winning Strategy Driven by Customer*
2 Devriendt, F., Berrevoets, J., & Verbeke, W. (2021). Why you should stop predicting customer churn and start using uplift models. *Information Sciences*, 548, 497–515.

PROJECT 1: SERVICE THAT STICKS



1 - Ribeiro, H., Barbosa, B., Moreira, A. C., & Rodrigues, R. G. (2024). Determinants of churn in telecommunication services: *Management Review Quarterly*, 74(3), 1327–1364. <https://doi.org/10.1007/s11301-023-00335-7>

2 - Vo, N. N. Y., Liu, S., Li, X., & Xu, G. (2021). Leveraging unstructured call log data for customer churn prediction. *Knowledge-Based Systems*, 212, 106586. <https://doi.org/10.1016/j.knsys.2020.106586>

3 - Ascarza, E. (2018). Retention Futility: Targeting High-Risk Customers Might be Ineffective. *Journal of Marketing Research*, 55(1), 80–98. <https://doi.org/10.1509/jmr.16.0163>

4 - Lemmens, A., & Gupta, S. (2020). Managing Churn to Maximize Profits. *Marketing Science*, 39(5), 956–973. <https://doi.org/10.1287/mksc.2020.1229>

RESULT 1: BETTER CHURN PREDICTION

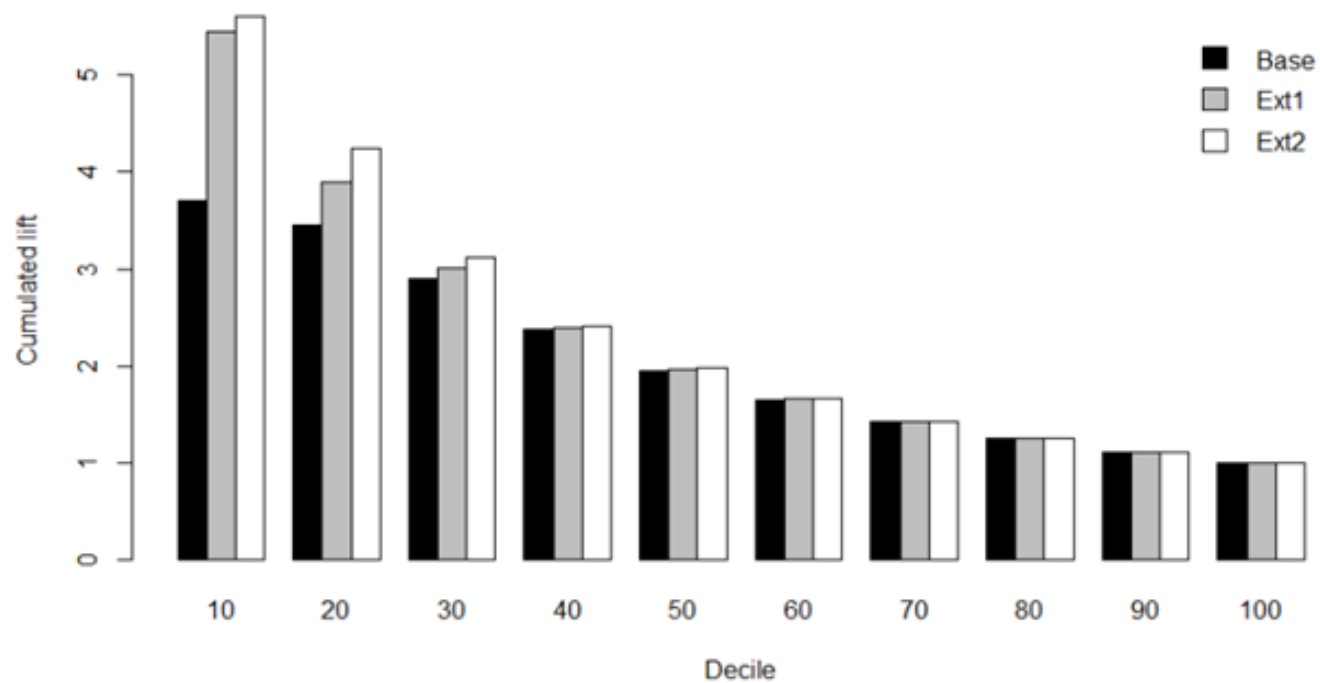


Table 15: Confusion matrix comparison

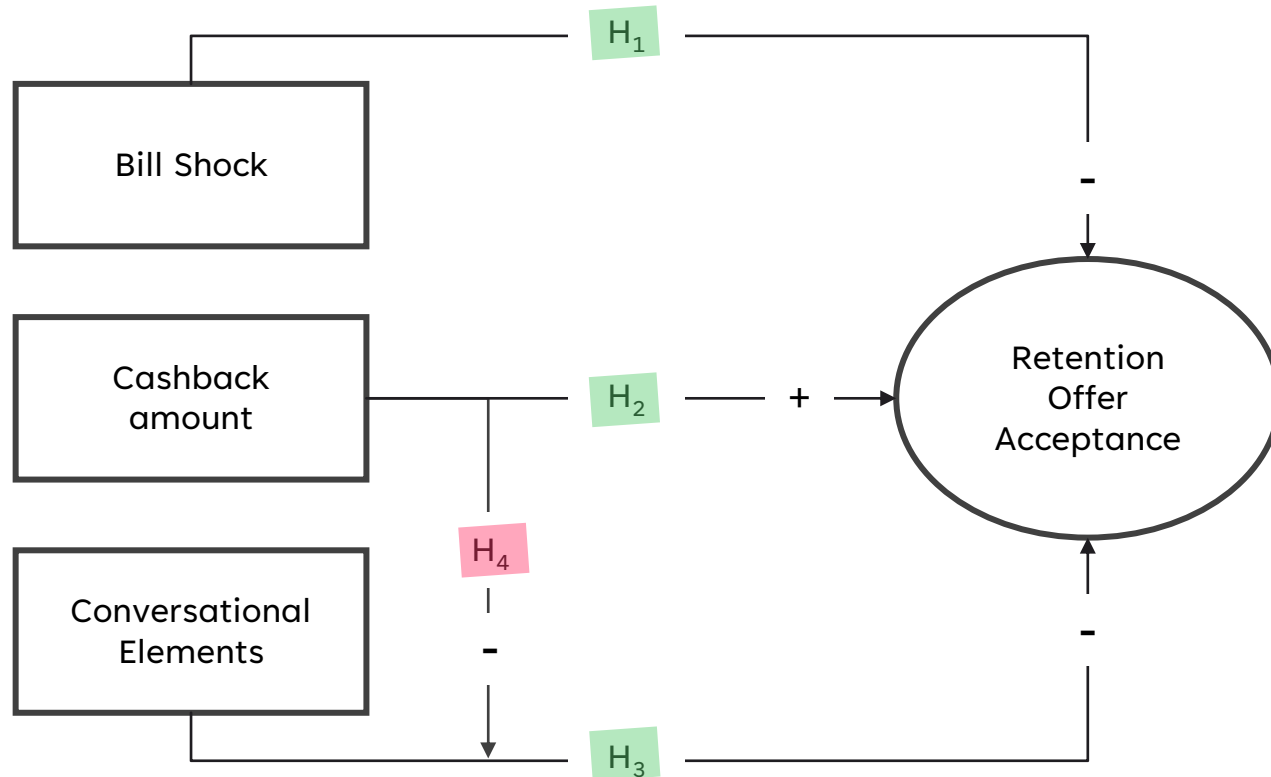
Predicted/ Actual	Base model		Extended model			
	True	False	Stage 1		Stage 2	
			True	False	True	False
True	4	267	28	247	38	237
False	14	3249	45	3217	58	3204

Table 16: AIC comparison between models

	Base Model	Extended Model	
		Stage 1	Stage 2
AIC	2192.4	1922.2	1990.7

Churn prediction has improved by ‘listening’ to what customers reveal in their customer service chats, using conversational elements. The extended models significantly outperformed the baseline model, with conversational elements contributing a 9× improvement in predictive sensitivity.

RESULT 2: CASHBACK OFFERS CAN'T FIX SUBPAR SERVICE



H₁ – Bill shock has a significant and negative impact on retention acceptance (OR=0.100 (±0.057), p<0.001)

H₂ – A high cashback amount has a significant and positive impact on retention acceptance (OR=2.900 (±1.936), p=0.001).

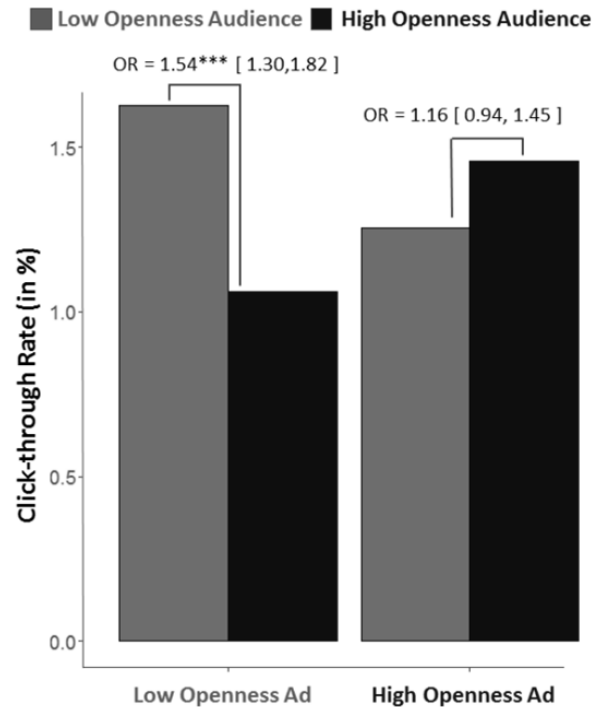
H₃ – Poor service quality has a significant and negative impact on retention acceptance (OR=0.161 (±0.094), p<0.001)

H₄ – The interaction between poor service quality and cashback amount is not statistically significant

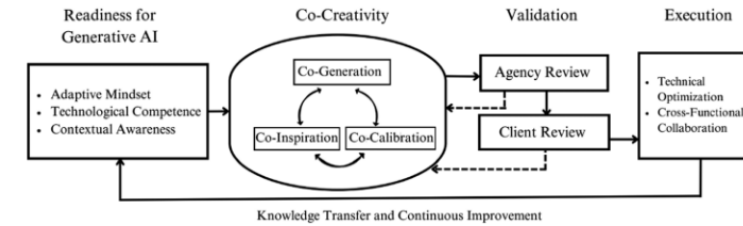
(OR = 0.953 (±0.660), p = 0.884), indicating that the negative effect of poor service on retention acceptance is not meaningfully altered by whether a generous cashback amount is offered.

All confidence intervals are at the 95% level

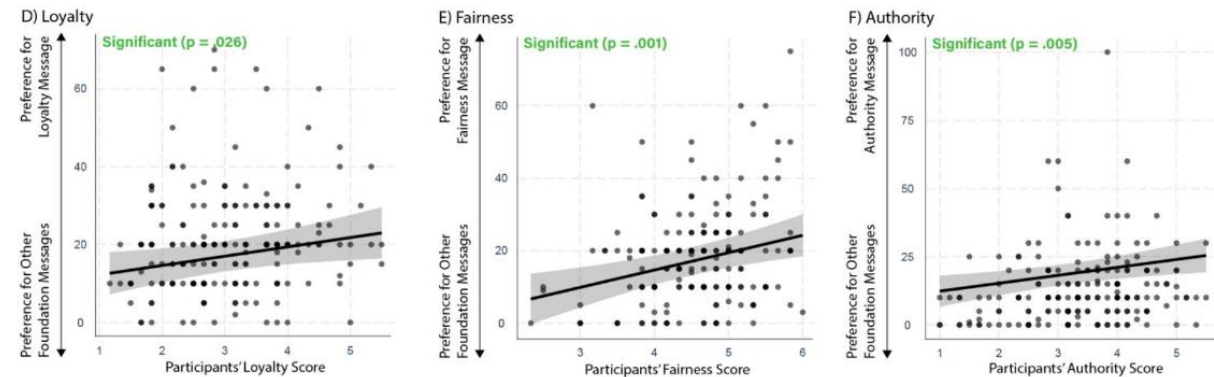
PROJECT 2: AI/HUMAN COLLABORATION IN CROSS SELL



Matching content with specific psychological profiles holds a promise of higher effectiveness, albeit sometimes asymmetrical¹.



AI should be considered a collaborative partner, rather than a disruptive threat².



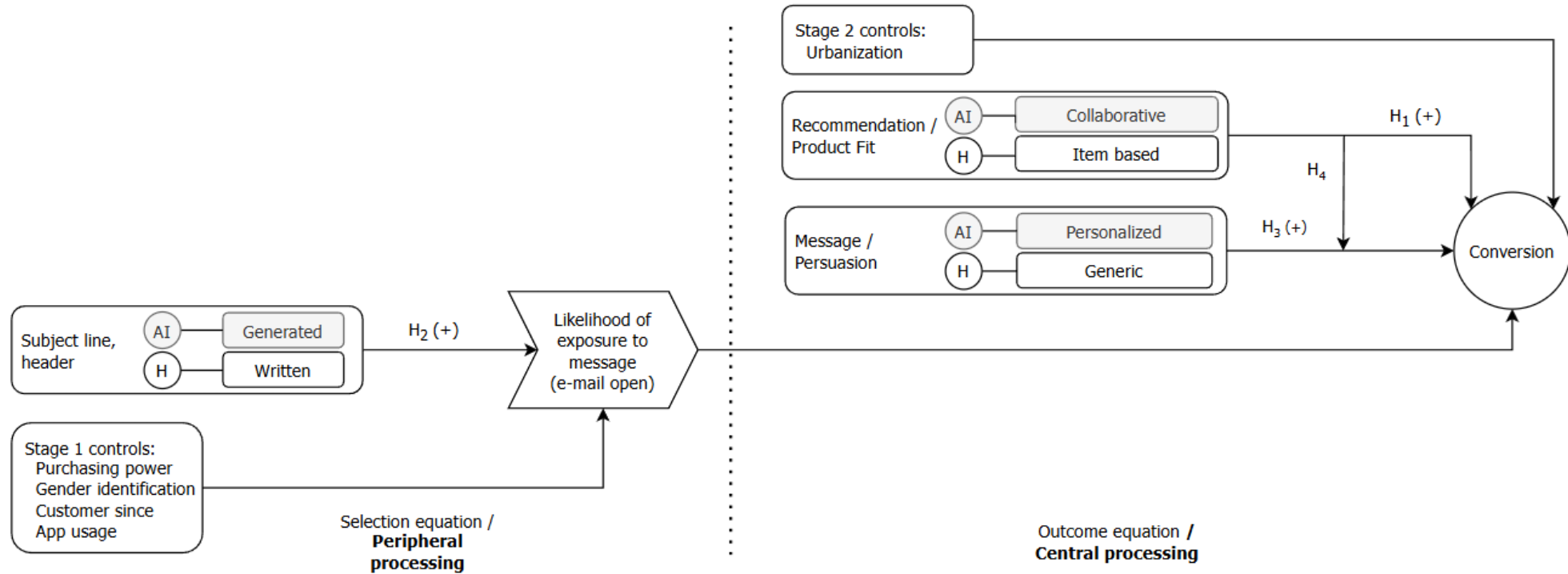
GenAI can craft personalized messages at scale that have positive influence on willingness-to-pay³.

1 - Matz, S. C., Kosinski, M., Nave, G., & Stillwell, D. J. (2017). Psychological targeting as an effective approach to digital mass persuasion. *Proceedings of the National Academy of Sciences*, 114(48), 12714–12719.

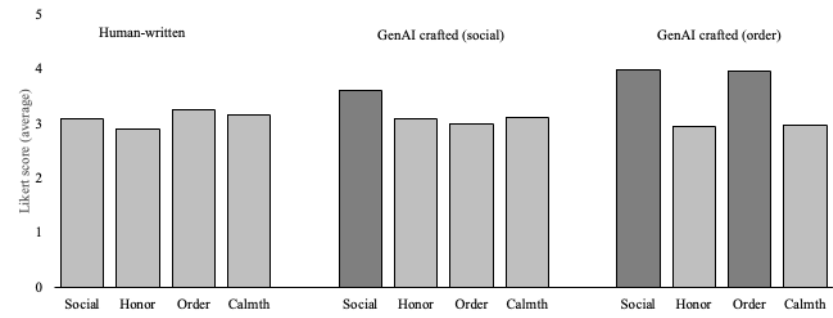
2 - Cui, W., Liu, M. J., & Yuan, R. (2025). Exploring the Integration of Generative AI in Advertising Agencies: A Co-Creative Process Model for Human–AI Collaboration. *Journal of Advertising Research*, 65(2), 167–189.

3 - Matz, S. C., Teeny, J. D., Vaid, S. S., Peters, H., Harari, G. M., & Cerf, M. (2024). The potential of generative AI for personalized persuasion at scale. *Scientific Reports*, 14(1), 4692.

PROJECT 2: AI/HUMAN COLLABORATION IN CROSS SELL



RESULTS: HUMAN AND AI ACTED AS COMPLEMENTS



Motives of the GenAI crafted content were recognized by participants in a pretest.

Hypothesis	Result	Support
H ₁ AI recommendations increase conversions	$\beta = -.007, p = .910$	No
H ₂ AI subject lines and headers increase opens	$\beta = .063, p = .006^{**}$	Yes
H ₃ AI personalized messages increase conversions	$\beta = -.106, p = .087^{\dagger}$	No
H ₄ AI recommendations paired with human messages increase conversions	$\beta = .181, p = .031^{*}$	Yes

Significance codes: $\dagger p < .100$, $^{*}p < .050$, $^{**}p < .010$, $^{***}p < .001$

We found partial support that dual processing was taking place and that GenAI crafted motive matched personalized subject lines and headers were able to capture attention.

Treatment combination		P(open)	P(conversion open)	P(open^conversion)
Recommendation	Message			
Human	AI	.482	.117	.056
Human	Human	.507	.147	.074
AI	AI	.501	.116	.058
AI	Human	.525	.222	.117
Average Interaction Effect		-.001	+.053*	+.018*
(Standard Error)		(.007)	(.023)	(.007)
95% Confidence Interval		[-.015,.013]	[.009,.098]	[.004,.033]

Note: Predictions evaluated at sample means of all covariates.

Average Interaction Effect computed as $[\Delta(\text{both}) - \Delta(\text{AI rec only})] - [\Delta(\text{human msg only}) - \Delta(\text{neither})]$.

Standard errors obtained via delta method. $N=11,976$; $n(\text{opened})=6,351$

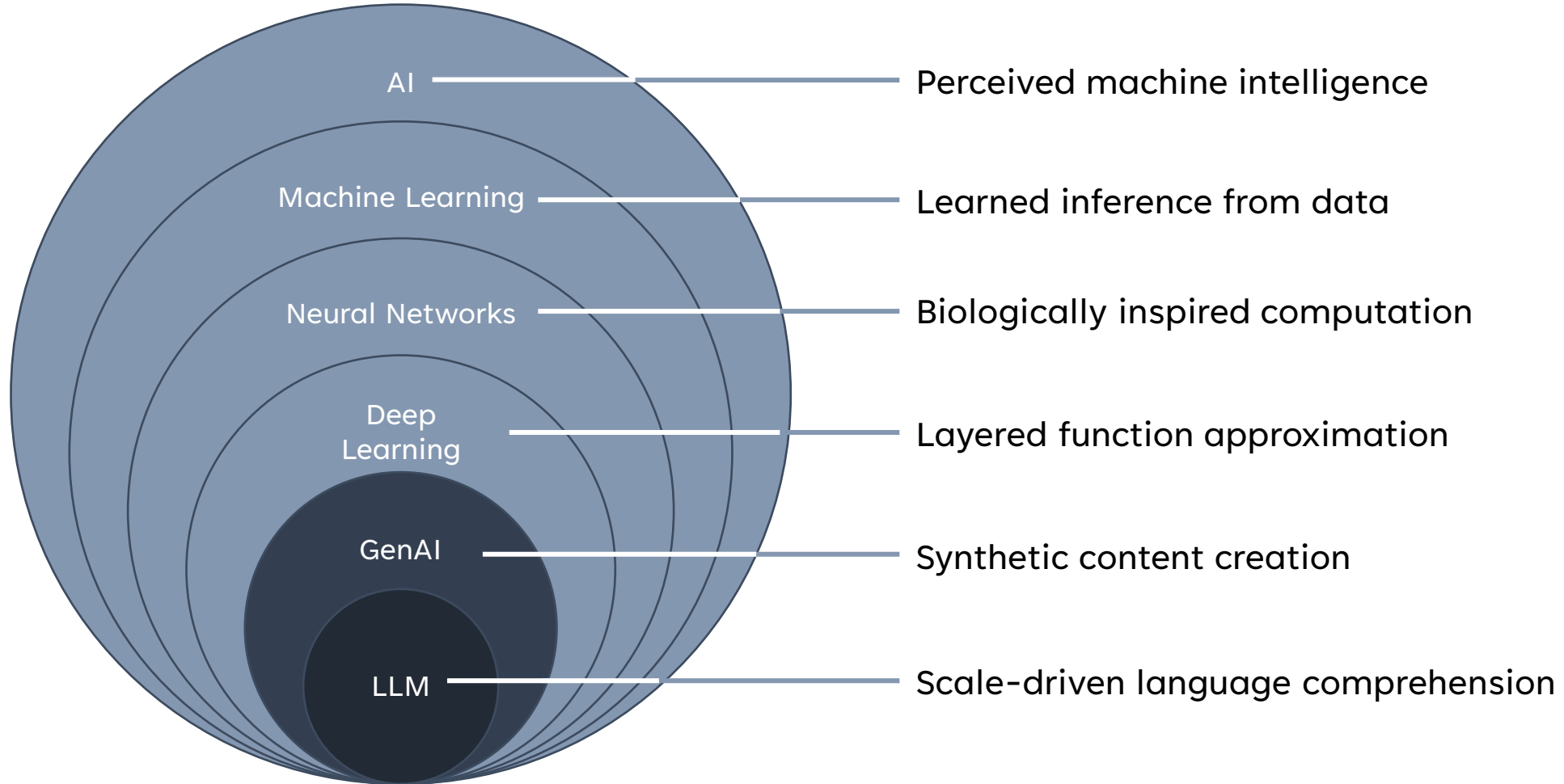
Significance codes: $\dagger p < .100$, $^{*}p < .050$, $^{**}p < .010$, $^{***}p < .001$

Our field experiment showed that a specific hybrid approach in Human-AI collaboration outperformed all other configurations: the marginal value of AI recommendations increases with the presence of human messaging.

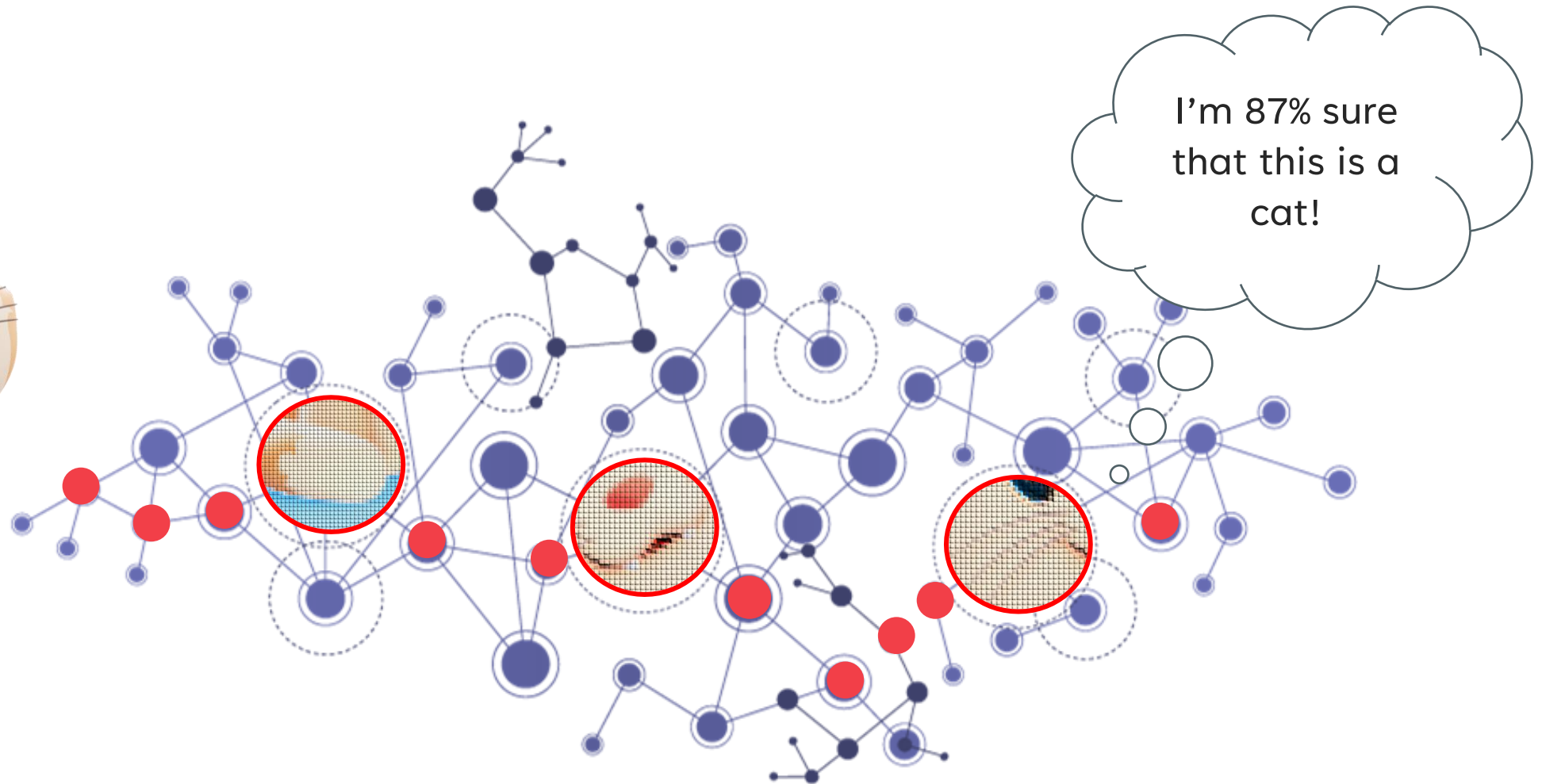


GENERATIVE AI THEORY

GENERATIVE AI AND LLMS IN THE BIGGER PICTURE



WHAT IS BIOLOGY INSPIRED COMPUTATION?



A self-organized network of neuron-like interlinked functions emerging from deep learning

HOW DID CHATGPT GET SO SMART?

It read everything there is on the Internet.
First step: **self-supervised pre-training**





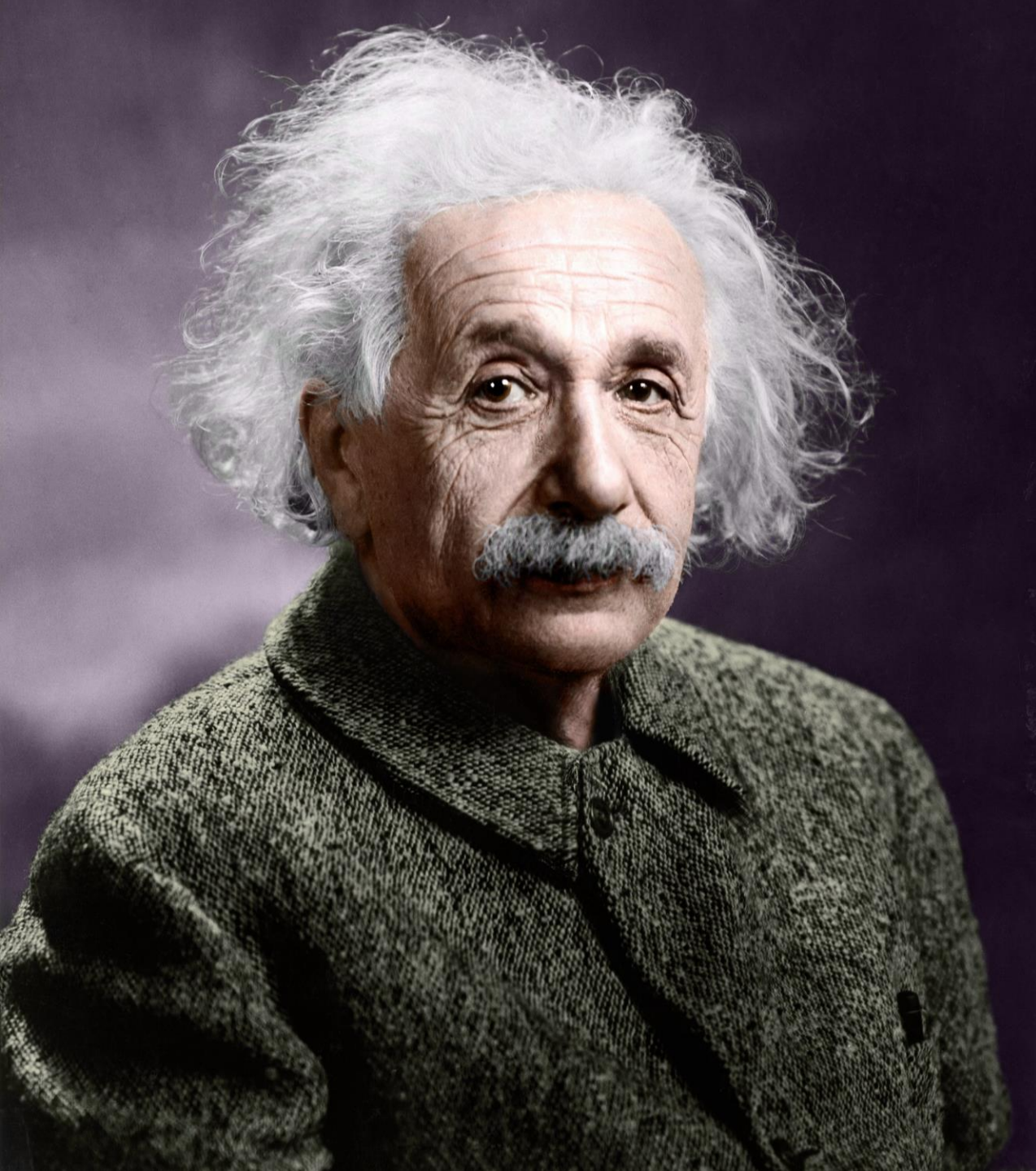
An apple a
day keeps the
doctor away



To be or not
to be , that is
the question.



May the force
be with you



The only thing
that interferes
with my
learning is my
teacher

AN LLM APPROACH TO THIS TASK: PRE-TRAINING

The only thing that interferes
with my learning is **a lack of motivation** (65%)

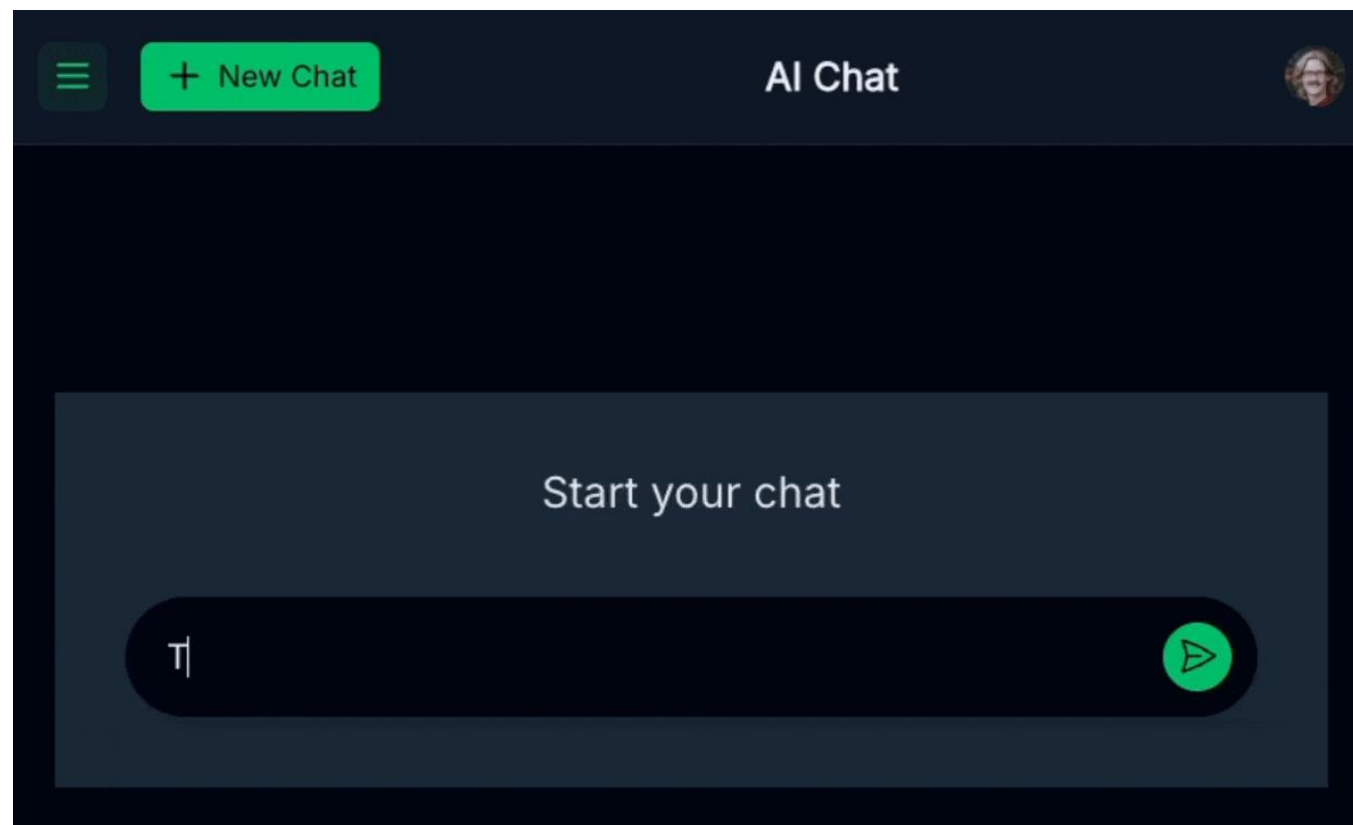
my crappy computer (61%)

my teacher (50%)

My dogs begging for attention (27%)

*This information is
learned from “the
Internet”* →

AFTER PRE-TRAINING: BASIC UNDERSTANDING OF NATURAL LANGUAGE



NEXT STAGE: SUPERVISED FINE-TUNING

Prompt

Reference answer

LLM answer

Explain photosynthesis in a few sentences

Photosynthesis is the process by which green plants, algae, and some bacteria convert light energy into chemical energy. Using sunlight captured by chlorophyll, they combine carbon dioxide from the air with water, producing glucose (a sugar that fuels growth) and releasing oxygen as a by-product. This reaction can be summarized as: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Plants are like tiny chefs that make their own food. They take **sunlight, water, and air (carbon dioxide)** and mix them together in their leaves. The sunlight is the “heat” that helps the mixing, and the result is a sugary food that gives the plant energy, while a gas called **oxygen** is puffed out into the air – the same oxygen we breathe! So, thanks to photosynthesis, plants stay healthy and give us fresh air.



What was Taylor Swift's debut album and how much did it sell?

Album: *Taylor Swift* (2006)
Sales: Roughly 7 million copies worldwide

Taylor Swift's first studio album, *Taylor Swift*, came out in late 2006 and has sold on the order of 8 million copies worldwide, making it a multi-platinum success and the launchpad for her career.




This information
comes from experts

NEXT STAGE: REINFORCEMENT LEARNING WITH HUMAN FEEDBACK



Understanding context: 'The question behind the question'.

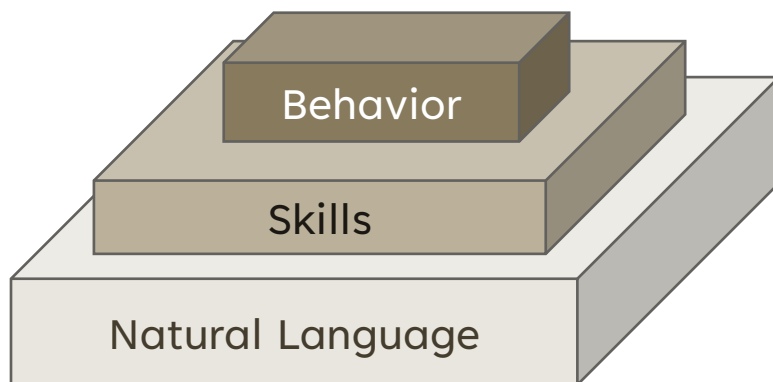
Bart Santing is a Dutch journalist, author, and media commentator who specializes in technology, cybersecurity, and privacy issues. Over the past two decades he has built a reputation as one of the Netherlands' most prominent voices on digital security, cyber-crime, and the societal impact of new technologies.

Weed out 'hallucinations'

I have no clue who **Bart Santing** is.

Reward preferred answers (A/B)

RESULT: MODEL THAT APPEARS TO REASON AND KNOW THINGS



codestral:latest

Yes, you can store the path to your local model as an environment variable and then access that variable in your Python script. This is a good approach if you want to make your code more flexible or if you need to run it on different machines with different file structures.

Here's how you can do it:

1. Set the environment variable `MODEL_PATH` to the path of your local model directory in your shell/terminal session (replace `"/path/to/your/local/model"` with the actual path):

For Unix-based systems (Linux, MacOS):

```
sh
1 export MODEL_PATH="/path/to/your/local/model"
```

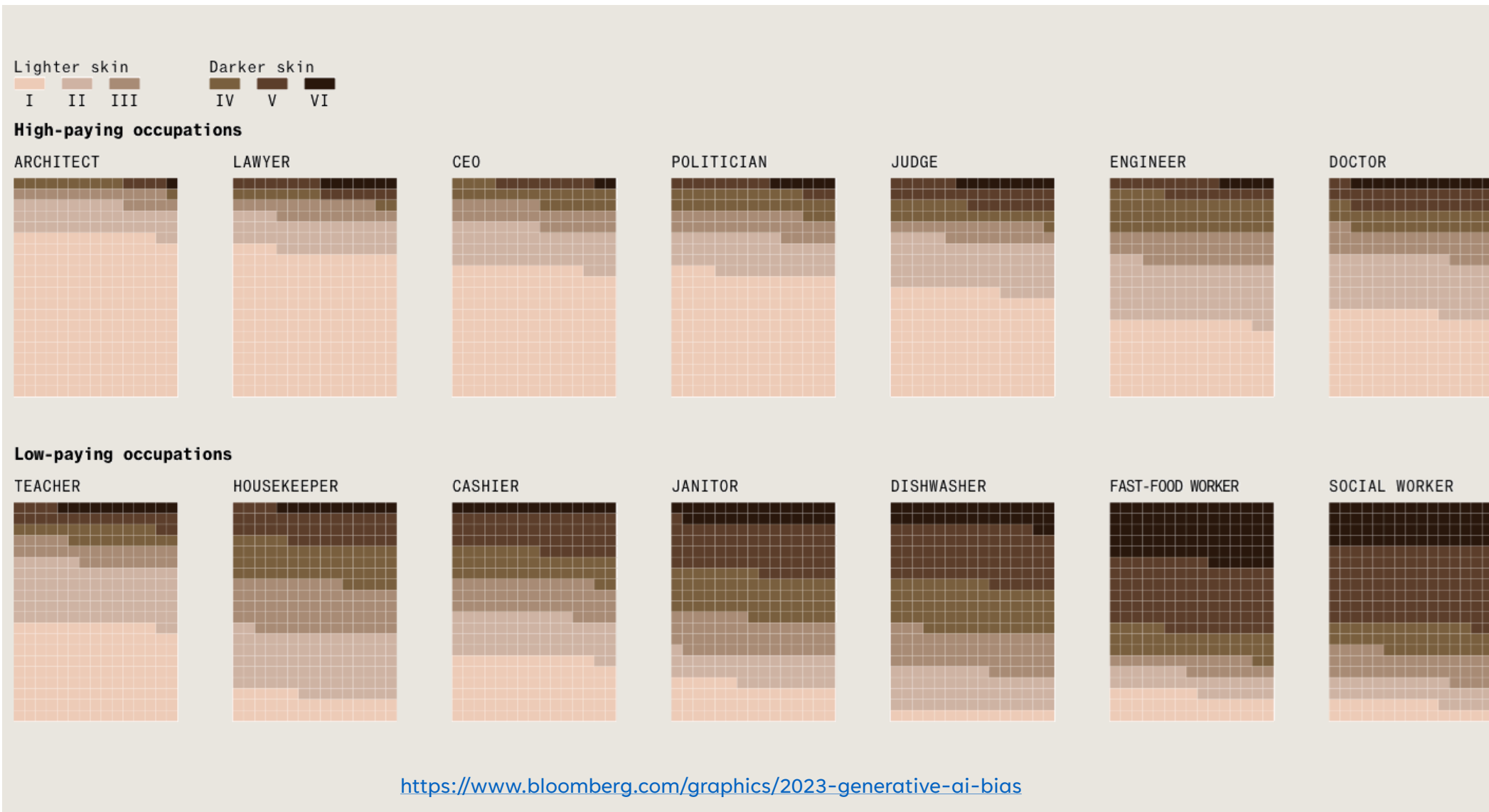
For Windows:

```
cmd
1 setx MODEL_PATH "/path/to/your/local/model"
```

2. In your Python script, use the `os` module to access the environment variable and load the model:

```
python
1 import os
2 from sentence_transformers import SentenceTransformer
3
4 # Get the path of the local directory where your model is saved from an environment variable
5 model_path = os.getenv('MODEL_PATH')
6
7 # Load the model
8 model = SentenceTransformer(model_path)
9
10 # Now you can use the model for inference
11 sentences = ["Hello, how are you?", "I'm fine, thank you."]
12 embeddings = model.encode(sentences)
```

INTERNALIZED BIAS: REGULARITIES, GAPS AND DISTORTIONS IN TRAINING DATA



INTERNALIZED BIAS: REGULARITIES, GAPS AND DISTORTIONS IN TRAINING DATA

Wanneer moet ik wel twee vakjes rood maken?

Perplexity

Als je op de combinatie van PvdA, GroenLinks stemt, dan moet je twee vakjes rood maken: één voor PvdA en één voor GroenLinks. Dat is de standaardprocedure bij gecombineerde stemprocedures. Heb je nog verdere vragen hierover?



You

I would like a photorealistic image of 4 salmons swimming in a river

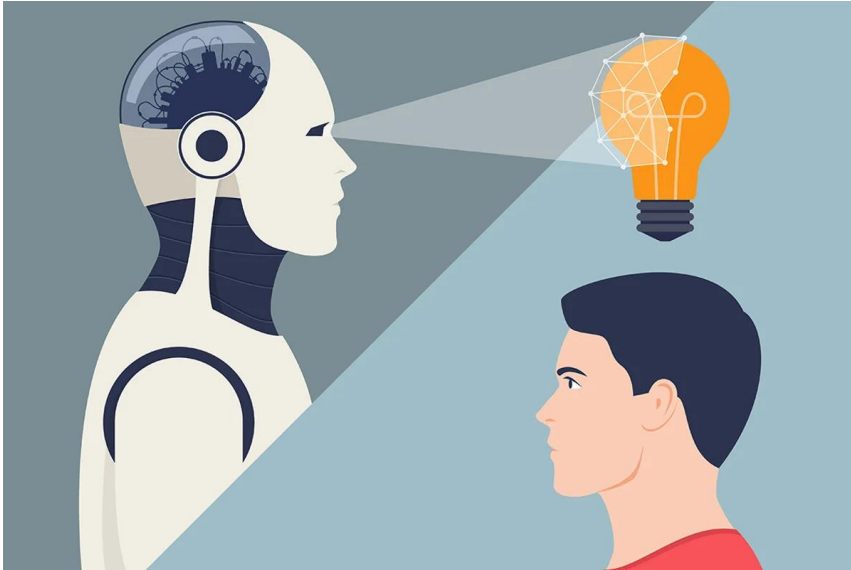


INTERACTION-BASED BIAS: THE ELIZA EFFECT AND ANTHROPOMORPHISM



<https://youtu.be/RMK9AphfLco>

AUTOMATION BIAS AND ALGORITHM AVERSION



Too much confidence in AI



Too little / no confidence in AI

EXTERNAL BIAS: FEEDBACK LOOP

I need a surprising pancake recipe

gpt-5

Here's a surprising, sweet-savory twist that still eats like classic pancakes.

Brown-Butter Miso Pancakes with Chili-Maple Sesame

Serves 2-3 (about 8 small pancakes)

Time: 25 minutes

Ingredients

- 1 cup (130 g) all-purpose flour
- 1 tbsp sugar
- 1 1/2 tsp baking powder
- 1/4 tsp baking soda
- 1 tbsp white miso paste (shiro miso)
- 1 cup plus 2 tbsp milk (dairy or oat)
- 1 large egg

gpt-4o

Certainly! How about trying Matcha Pancakes with a Citrus Glaze? The earthy flavor of matcha paired with a refreshing citrus glaze makes for a unique and delightful twist on traditional pancakes.

Matcha Pancakes with Citrus Glaze

Ingredients:

For the Pancakes:

- 1 cup all-purpose flour
- 1 tablespoon matcha green tea powder
- 2 tablespoons sugar
- 1 teaspoon baking powder
- 1/2 teaspoon baking soda
- 1/4 teaspoon salt



What would happen if the majority votes for matcha?

WHAT HAVE WE DISCUSSED?

Term	Plain English meaning
Deep learning	Training very large neural networks with many layers: the “depth of an LLM”
Self-supervised pre-training > Deep transformer LLM	The data itself provides the correct answer: “The LLM reads the whole internet and guesses missing words”
Supervised fine-tuning > fine-tuned LLM	Humans provide the correct answer a prior: “Humans teach the LLM a specific skill”
RLHF > aligned LLM	Rewards and penalties given, based on outcome: ”Humans make the LLM behave the way humans like it to”
Internalized (model intrinsic) bias	Regularities, gaps, or distortions in the data become statistically imprinted in the LLM: stereotypes, ideology, toxicity.
External (interaction level) bias	Emerges from the human-machine interface, downstream pipelines, or the broader socio-technical ecosystem: automation bias/algorithm aversion, feedback-loop bias.



GENERATIVE AI PRACTICE

HOW HARD CAN IT BE TO TELL A MODEL WHAT TO DO?



<https://youtu.be/FN2RM-CHkuI>

PROMPTING BEST PRACTICES: PUT YOURSELF IN THE LOOP

Role: You are a [role].

Task: Do [task] for [audience/use case].

Context: [BEGIN CONTEXT] ... [END CONTEXT]

Constraints: [tone], [length], [domain rules], [time horizon], [do/don't].

Output format: Return JSON only with these keys and types.

Quality standard: Must satisfy [criteria]. If uncertain or unsupported, say so and explain what's needed.

Examples:

• Input: [mini-input] → Output: [mini-output]

Now complete the task for: [current input].

MULTI-SHOT PROMPTING: CORRECTING EARLIER GAPS

Could you write a blog about dental healthcare?

Smile Bright: The Essentials of Modern Dental Healthcare

When it comes to overall health, your mouth is often the first-place problems surface—so a solid dental routine isn't just about a pretty smile; it's a cornerstone of well-being. Here's a lightning-fast guide to staying on top of your teeth and gums.

Can you change the angle a bit more towards healthcare insurance?

Smile-Savvy: Why Dental Insurance Is the Real MVP of Oral Health

Your teeth don't just need a brush and floss—they need a safety net. A good dental insurance plan turns routine care into a cost-free habit and shields you from pricey emergencies. Here's a bite-size guide to making that net work for you.

WHEN AUTOMATED: CHAIN-OF-PROMPTS

ADVANCED PROMPTING TECHNIQUES ARE GETTING OUT OF FASHION

Can you form a panel of experts and discuss the benefits of healthcare insurance?

Prof. Luis Ortega (Economist)

"Health insurance is a risk-pooling mechanism. By spreading medical costs across many, it eliminates the catastrophic financial risk that can derail a household's finances. Economically, it supports labor mobility, entrepreneurship, and overall consumer confidence."

Linda Wu (Actuary)

"From a financial perspective, insurance provides predictability. It caps out-of-pocket spending, allowing households to budget effectively. It also funds preventive services that reduce long-term claims, benefiting both insurers and patients."

← This is a Tree-of-Thoughts (ToT) prompt



This is because the de facto standard in modern LLMs has become: **reasoning / thinking** and because of something else...

SUPPOSE OHRA WOULD HAVE ROLLED OUT CHATGPT-3.5 TO CUSTOMERS

System Prompt:

You are a helpful and friendly assistant for OHRA customers. You are knowledgeable in car insurance and you're eager to answer any related questions.

Chat conversation (user prompts):

Does my policy cover windshield damage?

Not specific

Well.. that depends on your coverage and the specific circumstances under which it got damaged.

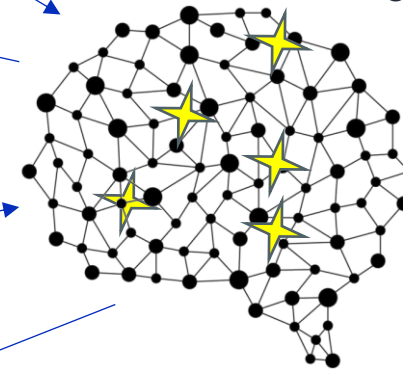
Can I get a discount on my second car insurance policy?

False

Unfortunately, OHRA doesn't provide volume discounts on insurance policies

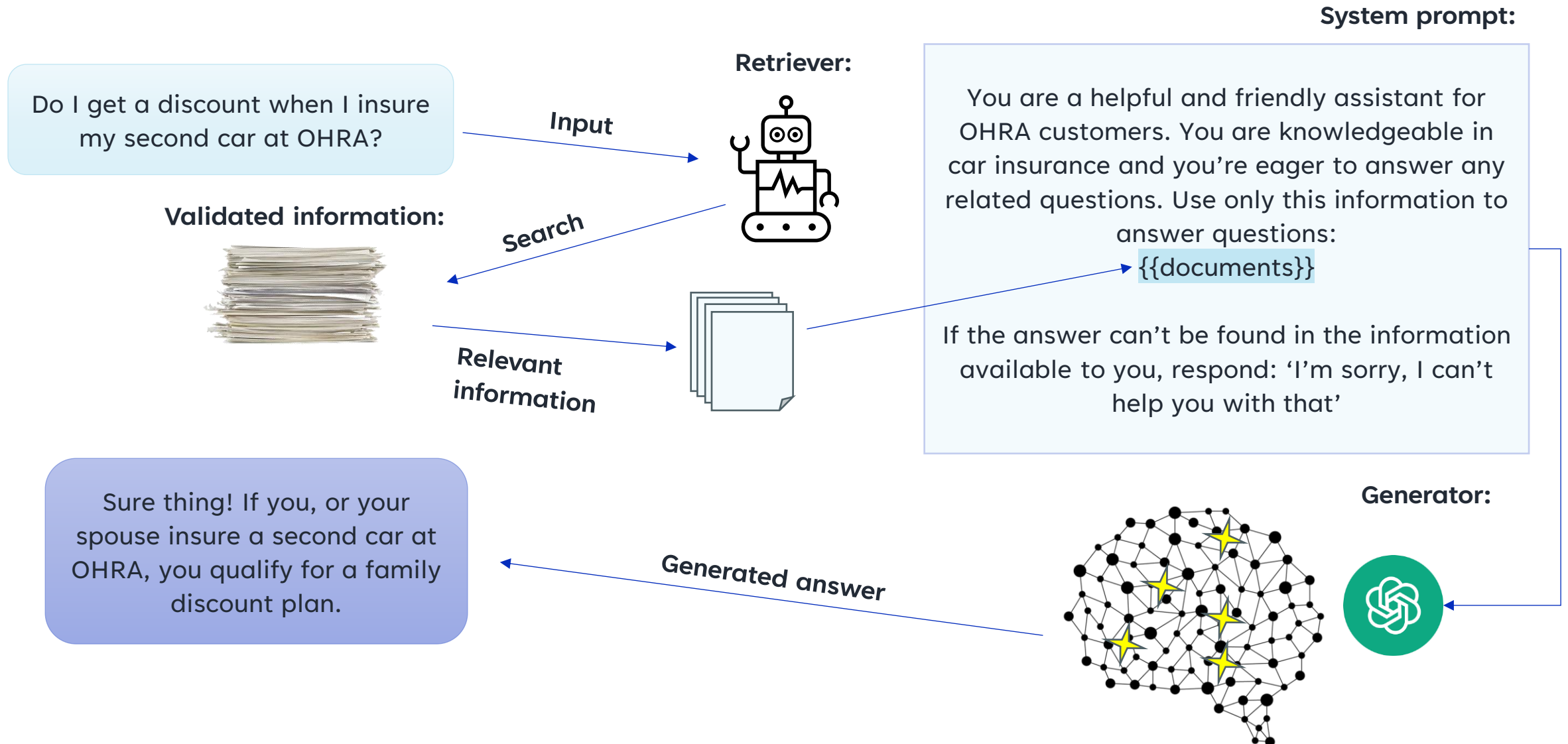
Input

Generator:

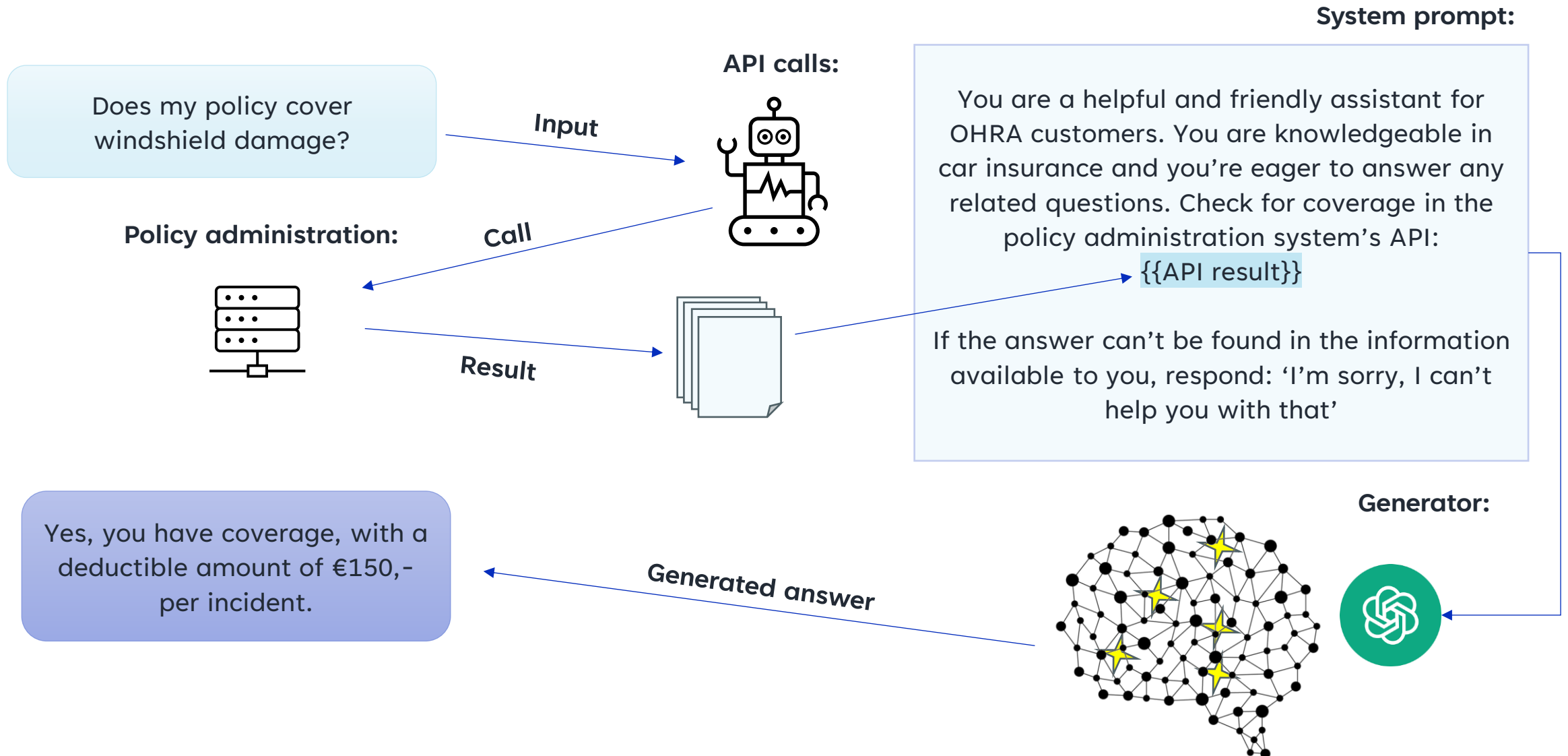


Generated answer

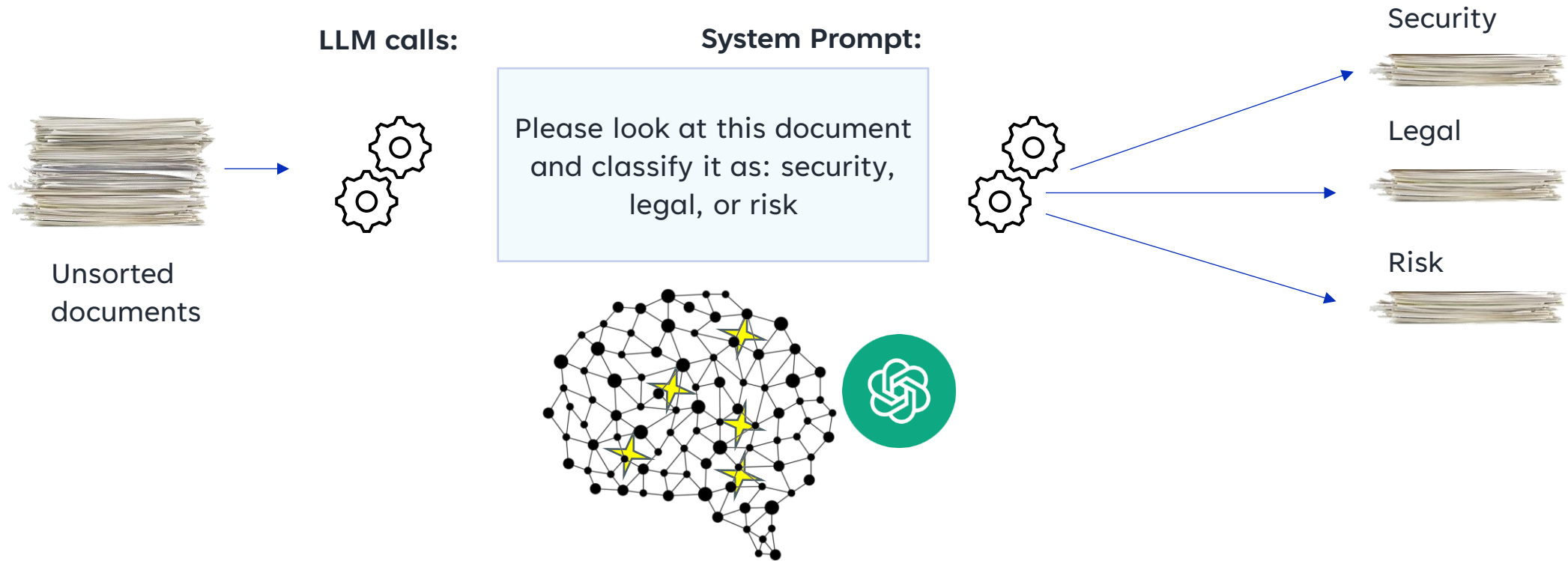
INSTEAD, **RAG** IS BEING USED TO GENERATE BETTER ANSWERS



TOOLS CAN BE HELPFUL TO GET RELEVANT INFORMATION IN CONTEXT



AUTOMATION TAKES THE HASSLE OUT OF REPETITIVE TASKS

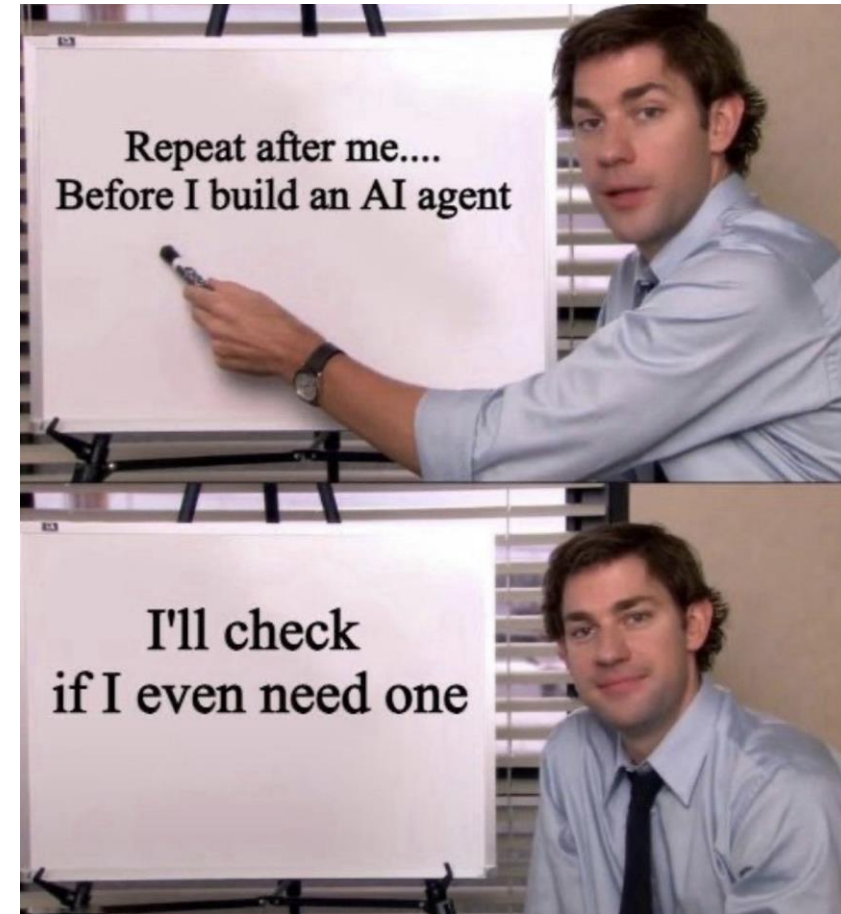


In this example, the LLM is only **part** of the automation flow, not the main component.

THE TERM 'AGENT' IS APPLIED LOOSELY IN THE CORPORATE WORLD

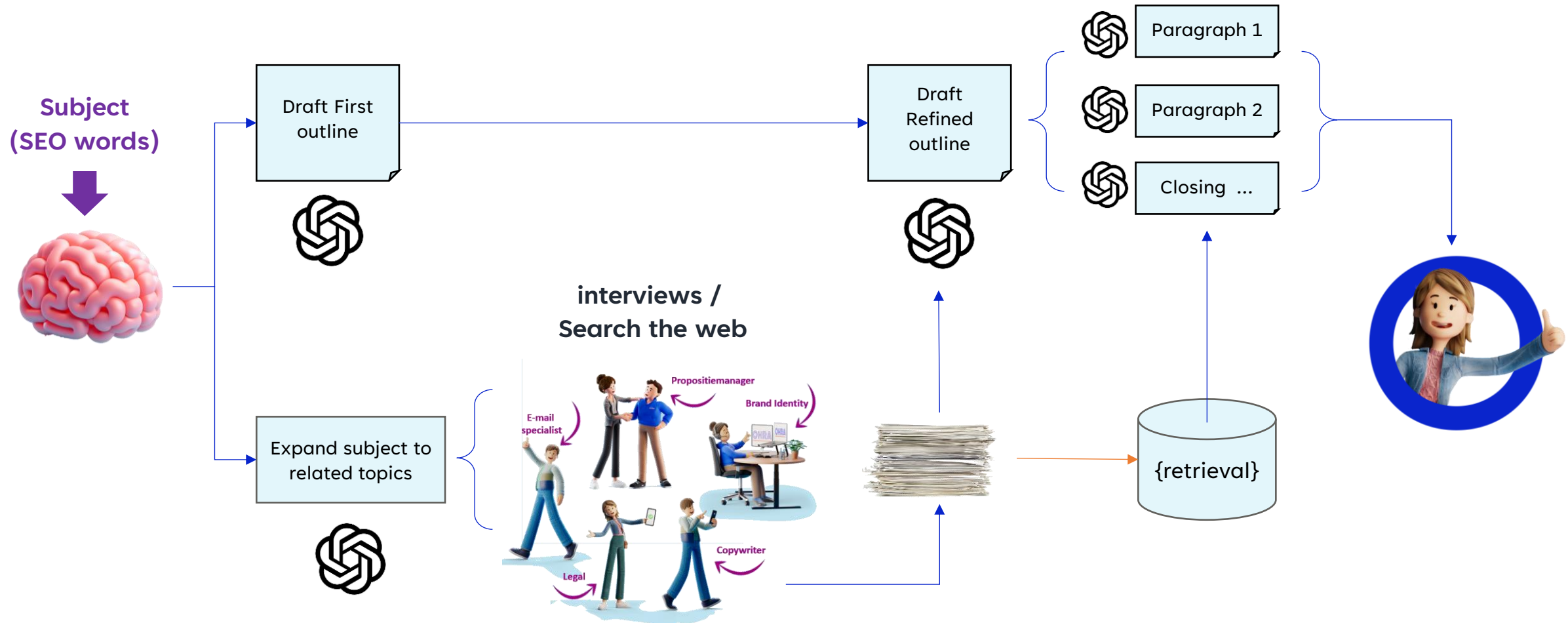


Agents should be embedded in the models or governed through a framework.



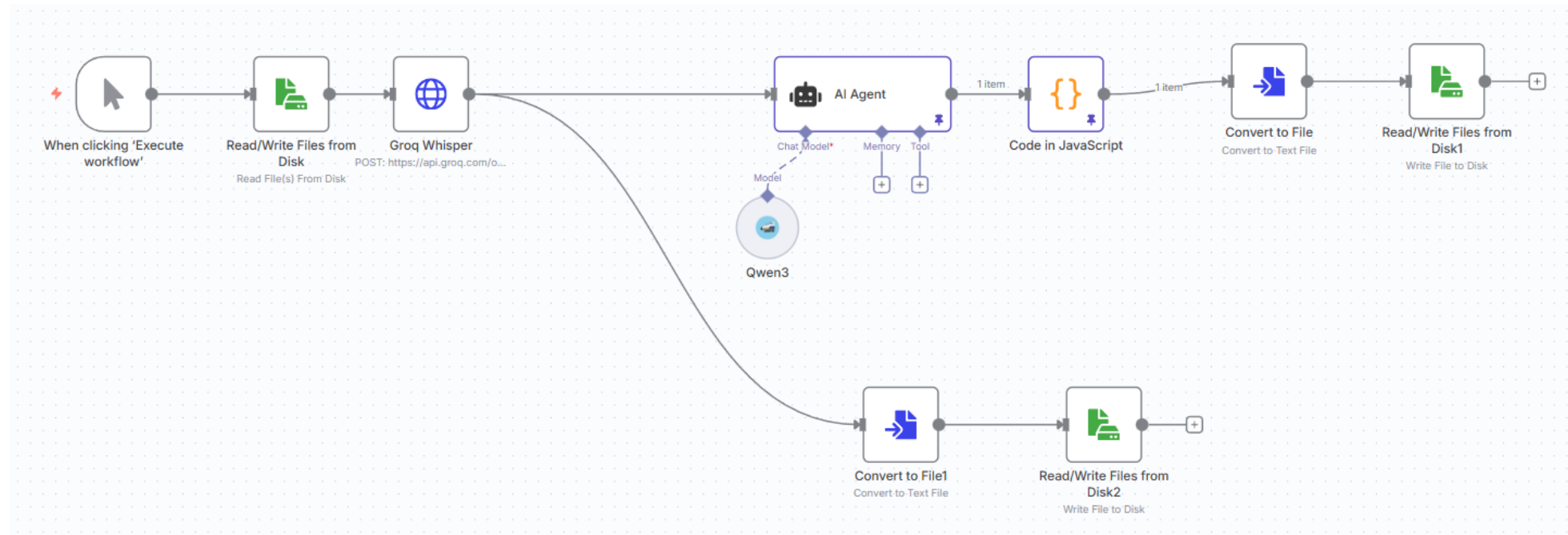
Agents are no silver bullet.
Sometimes other solutions work better.

OHRA EXAMPLE: AGENTIC CONTENT CREATION FRAMEWORK



³ Shao, et al (2024) <https://doi.org/10.48550/arXiv.2402.14207>

AND THERE'S LOTS YOU CAN DO IN YOUR OWN HOMELAB TOO!



A series of white, overlapping geometric lines on a black background, creating a complex, abstract pattern on the left side of the slide.

THANK YOU

Bart Santing

b.c.santing@uva.nl

